

Cleverscope CS548 Performance Verification Specification

Summary

The Cleverscope CS548 includes internal resources to calibrate an internal voltage reference against an external voltage reference (External Once Yearly Calibration), and the ability to calibrate the two voltage ranges ($\pm 8V$ and $\pm 0.8V$) against an internal reference (Standard Internal Calibration).

In addition the CS548 provides a high accuracy three level voltage reference (0, 681.8, 7500 mV) which can be used as the external reference if required (in versions 1.3 onwards).

The purpose of this test specification is to verify the performance of the internal calibration system, and to verify that the timing system is in specification.

Three systems need to be verified - the internal voltage reference used to calibrate the internal ranges against, the two ranges, and the timing system. The number of measurements needed to make this verification are relatively low.

Cleverscope recommends that users perform an internal calibration (Settings/Calibrate Cleverscope /Standard Internal Calibration) when they want to be assured of correct performance. The Cleverscope application also provides the ability to calibrate the internal reference against an external reference, but this is only required very occasionally (eg yearly), as the internal reference is temperature stabilized, and once calibrated will provide good long term stability. A built in high accuracy reference is available via the *Cal Ref* output on the left hand side of the CS548 front panel.

Equipment

You will need the following equipment and software:

1. Cleverscope CS548 with power supply, USB cable, and two scope probes.
2. A voltage source and calibrated voltmeter, or a calibrated voltage source, or the CS548 Cal Ref.
3. A signal generator and calibrated frequency counter, or a calibrated signal source and BNC cable.
4. PC loaded with the Cleverscope application.

Amplitude Accuracy

The specified amplitude accuracy is:

- $\pm 8V$ Range: $\pm 0.15\% \pm 8mV$
- $\pm 0.8V$ Range: $\pm 0.15\% \pm 800\mu V$
- Pulse response flatness: 20n - 5us <2%, 5us+ <0.2%

Frequency Accuracy

The specified frequency accuracy is:

- ± 1 ppm

Procedure

The verification procedure is:

1. Connect up the Cleverscope CS548 acquisition unit, and use Auto to verify communication.
2. Leave the acquisition unit to warm up for at least 20 minutes. This allows the digitizer temperature system to stabilize at 40 deg C.
3. Using the menu Settings/Calibrate Cleverscope to open the calibration dialog, choose the Self Cal tab, ensure all cables disconnected from the front panel, and click on 'Start Self Calibration' . Wait until complete, and then click 'Finish'.
4. Set up the oscilloscope with the supplied probes connected Channels A - D, all probes and ground clips connected to the known voltage source. The probes will be switched between x1 and REF during the test. You can do one channel at a time, or all channels at the same time.

5. DC Voltage measurements

Using the table below, open the *CS548 Calibration.apc* file, and click Auto. Set the voltage source approximately to the specified voltage. If using a voltage source and calibrated voltmeter, write down the reference value (the reference value will be the same as the specified value when using a calibrated voltage source). Write it down with at least three digits precision. Set the probe switches as specified, and using the Signal Information display, record the DC voltage. Write down the difference between the measured value and the reference value (dA.). If the difference is less than given, tick the passed box. Do this for all channels. If using the DC Cal source, reverse the probe direction to get negative voltages. Verify the DC Cal output before starting with a 6.5 digit voltmeter or better. Averaging is used for maximum accuracy. Click F1 and F2 for the correct setup as shown in the table. Set the Vcal value as required - either on the SG control panel or with an external source.

Range Values: Chan A (See following pages for Chan A-D)

These voltages are available from the CS548 Cal ref Output or an External Source:

File: *CS548 Calibration.apc*

Scope Graph Preset (F1 or F1)	Range	Vs. Specified Value (V).	Vr. Ref Value.	dVmax. Max delta value (mV).	Probe setting.	mV DC value	delta value	Pass? delta <= dVmax?
F1	8V	0	0	8	Ref			
F2	8V	7.5 V		19	X1			
F2	8V	-7.5 V		19	X1			
F2	8V	681.8mV		9				
F2	8V	-681.8mV		9				
F1	0.8V	0	0	0.8	Ref			
F1	0.8V	681.8 mV		1.8	X1			
F1	0.8V	-681.8 mV		1.8	X1			

6. After having measured both the +V and -V values, calculate the difference between V+ and V-, dV. Verify that dV falls in the range given.

Difference Values – A Channel

Vs Specified value	mVa ⁺	mVa ⁻	Delta mVa ⁺ - mVa ⁻	Min Delta (mV)	Max Delta (mV)	Pass?
7.5V				14,977	15,023	
681.8 mV				1,361	1,367	

7. Pulse Flatness

File: *CS548 Calibration.apc*

Load the file, and plug a scope probe into the Probe Comp output or a calibrated 0 - 7.5V 1 kHz square wave source. The source flatness must be <0.1%. Set the probe switch to 10x. Select the trigger source to the current channel, and plug the probe into the channel to be measured. The square wave should be visible on the screen. For maximum accuracy click Do Averaging to ON on the Cscope Control panel.

Adjust the probe trim with a non metallic adjustment tool to achieve maximum flatness as displayed on the Tracking Display. Do for both the $\pm 0.8V$ and $\pm 8V$ Ranges, by selecting first the F1 button on the Scope Display ($\pm 0.8V$) or the F2 button ($\pm 8V$). Place Markers as shown at about 20ns and 450us, and 5us and 450us, by clicking '1' to place the left hand marker and '2' to place the right hand marker. Use the Tracer value to see the time, and observe the dV value:

F1 ($\pm 0.8V$) Range



The limits are: 20n - 5us <2%, 5us+ <0.2%

Pulse Flatness

7.5V 1 kHz square wave, x10 probe setting, displayed as 750 mV (approx)	Peak - Marker 2 (mV)	Peak Limit (mV)	Marker 2 - Marker 1 (mV)	Flat Limit (mV)	Pass?
$\pm 0.8V$ - F1 on Scope Graph		15		1.5	
$\pm 8V$ - F2 on Scope Graph		150		15	

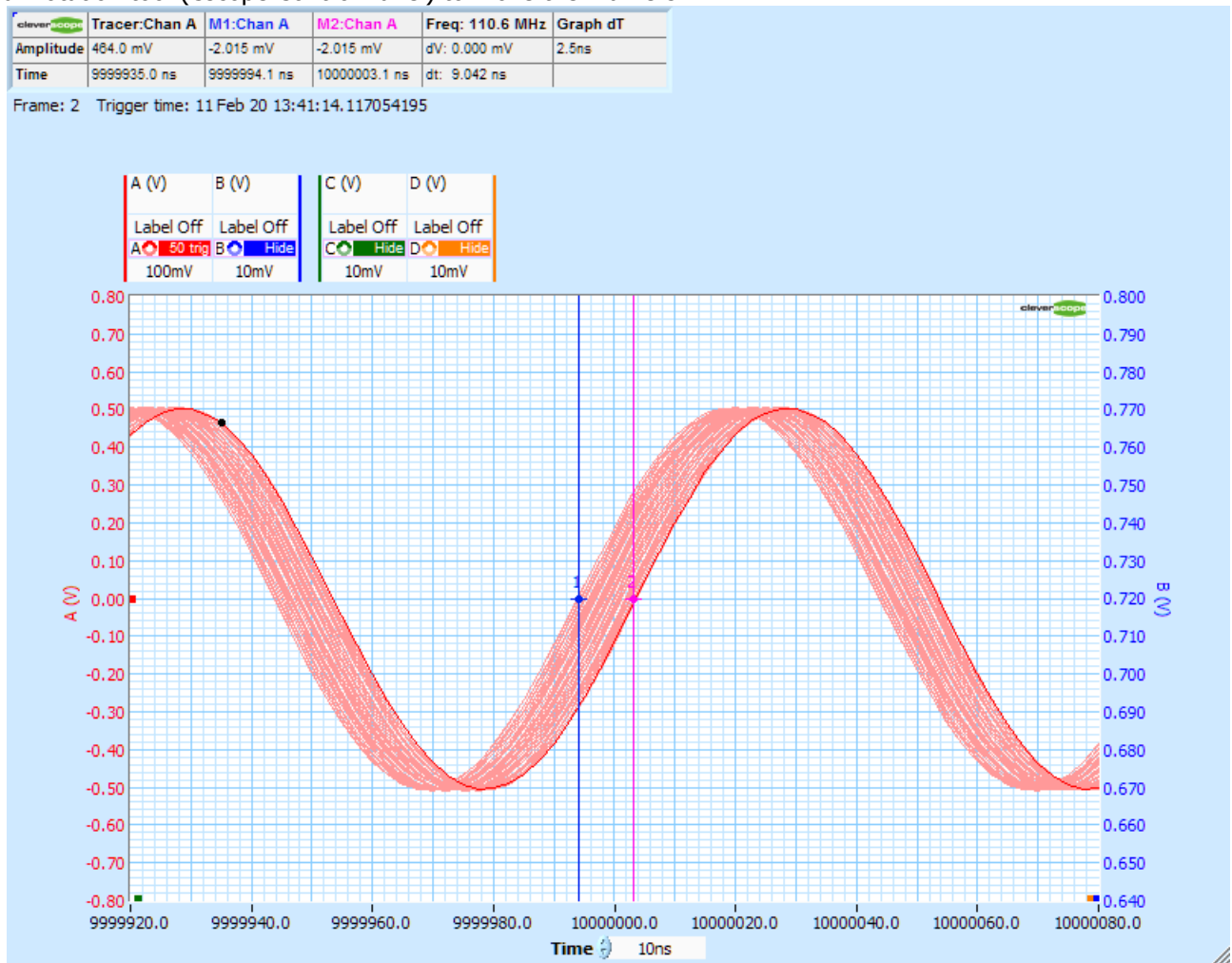
8. Timebase Accuracy

File: CS548 Timebase.apc

Use a calibrated signal generator outputting a 100 Hz square wave, 1V p-p, centred on 0V, connected via BNC to channel A. The 100 Hz square wave frequency accuracy must be less than 0.1ppm. Load the file. Click on 'Triggered'. Observe the rising edge at 10 000 000ns on the Tracking graph. 1 division (10ns) corresponds to 1 ppm.

Alternately use a 10 MHz sinusoidal reference source, and observe the waveform at 10 000 000 ns with 10ns per division. There will be jitter in the sampling position, because trigger interpolation is not used for such a long time period. Ensure the total jitter is less than +/- 1ppm, and centred on 10 000 000 ns. Here is an example capture with a 1ppm 10 MHz reference, and persistence set to 50 captures, displaying a peak variation of 9ns, centred on 10ms, or $\pm 0.45\text{ppm}$:

You will need to reposition the markers by double clicking on the edges of the waveform. Use the annotation tool (Cscope Control Panel) to move the markers.



☐ Tick here if the variation from the 10 000 000ns is less than ± 1 divisions.

All Channel Results

Serial Number:

Chan A

Measured Values

Scope Graph Preset (F1 or F2)	Range	Vs. Specified Value (V).	Vr. Ref Value.	dVmax. Max delta value (mV).	Probe setting.	mV DC value	delta value	Pass? delta <= dVmax?
F2	8V	0	0	8	Ref			
F2	8V	7.5 V		19	X1			
F2	8V	-7.5 V		19	X1			
F2	8V	681.8mV		9				
F2	8V	-681.8mV		9				
F1	0.8V	0	0	0.8	Ref			
F1	0.8V	681.8 mV		1.8	X1			
F1	0.8V	-681.8 mV		1.8	X1			

Difference Values

Vs Specified value	mVa ⁺	mVa ⁻	Delta Va ⁺ - Va ⁻	Min Delta (mV)	Max Delta (mV)	Pass?
7.5V				14,977	15,023	
681.8 mV				1,361	1,367	

Pulse Flatness

7.5V 1 kHz square wave, x10 probe setting, displayed as 750 mV (approx)	Peak - Marker 2 (mV)	Peak Limit (mV)	Marker 2 - Marker 1 (mV)	Flat Limit (mV)	Pass?
±0.8V - F1 on Scope Graph		15		1.5	
±8V - F2 on Scope Graph		150		15	

Timebase Accuracy

☐ Tick here if the variation from the 10 000 000ns is less than ±1 divisions.

Chan B

Measured Values

Scope Graph Preset (F1 or F2)	Range	Vs. Specified Value (V).	Vr. Ref Value.	dVmax. Max delta value (mV).	Probe setting.	mV DC value	delta value	Pass? delta <= dVmax?
F2	8V	0	0	8	Ref			
F2	8V	7.5 V		19	X1			
F2	8V	-7.5 V		19	X1			
F2	8V	681.8mV		9				
F2	8V	-681.8mV		9				
F1	0.8V	0	0	0.8	Ref			
F1	0.8V	681.8 mV		1.8	X1			
F1	0.8V	-681.8 mV		1.8	X1			

Difference Values

Vs Specified value	mVa ⁺	mVa ⁻	Delta Va ⁺ - Va ⁻	Min Delta (mV)	Max Delta (mV)	Pass?
7.5V				14,977	15,023	
681.8 mV				1,361	1,367	

Pulse Flatness

7.5V 1 kHz square wave, x10 probe setting, displayed as 750 mV (approx)	Peak - Marker 2 (mV)	Peak Limit (mV)	Marker 2 - Marker 1 (mV)	Flat Limit (mV)	Pass?
±0.8V - F1 on Scope Graph		15		1.5	
±8V - F2 on Scope Graph		150		15	

Timebase Accuracy

- ☐ Tick here if the variation from the 10 000 000ns is less than ±1 divisions.

Chan C

Measured Values

Scope Graph Preset (F1 or F2)	Range	Vs. Specified Value (V).	Vr. Ref Value.	dVmax. Max delta value (mV).	Probe setting.	mV DC value	delta value	Pass? delta <= dVmax?
F2	8V	0	0	8	Ref			
F2	8V	7.5 V		19	X1			
F2	8V	-7.5 V		19	X1			
F2	8V	681.8mV		9				
F2	8V	-681.8mV		9				
F1	0.8V	0	0	0.8	Ref			
F1	0.8V	681.8 mV		1.8	X1			
F1	0.8V	-681.8 mV		1.8	X1			

Difference Values

Vs Specified value	mVa ⁺	mVa ⁻	Delta Va ⁺ - Va ⁻	Min Delta (mV)	Max Delta (mV)	Pass?
7.5V				14,977	15,023	
681.8 mV				1,361	1,367	

Pulse Flatness

7.5V 1 kHz square wave, x10 probe setting, displayed as 750 mV (approx)	Peak - Marker 2 (mV)	Peak Limit (mV)	Marker 2 - Marker 1 (mV)	Flat Limit (mV)	Pass?
±0.8V - F1 on Scope Graph		15		1.5	
±8V - F2 on Scope Graph		150		15	

Timebase Accuracy

- ☐ Tick here if the variation from the 10 000 000ns is less than ±1 divisions.

Chan D

Measured Values

Scope Graph Preset (F1 or F2)	Range	Vs. Specified Value (V).	Vr. Ref Value.	dVmax. Max delta value (mV).	Probe setting.	mV DC value	delta value	Pass? delta <= dVmax?
F2	8V	0	0	8	Ref			
F2	8V	7.5 V		19	X1			
F2	8V	-7.5 V		19	X1			
F2	8V	681.8mV		9				
F2	8V	-681.8mV		9				
F1	0.8V	0	0	0.8	Ref			
F1	0.8V	681.8 mV		1.8	X1			
F1	0.8V	-681.8 mV		1.8	X1			

Difference Values

Vs Specified value	mVa ⁺	mVa ⁻	Delta Va ⁺ - Va ⁻	Min Delta (mV)	Max Delta (mV)	Pass?
7.5V				14,977	15,023	
681.8 mV				1,361	1,367	

Pulse Flatness

7.5V 1 kHz square wave, x10 probe setting, displayed as 750 mV (approx)	Peak - Marker 2 (mV)	Peak Limit (mV)	Marker 2 - Marker 1 (mV)	Flat Limit (mV)	Pass?
±0.8V - F1 on Scope Graph		15		1.5	
±8V - F2 on Scope Graph		150		15	

Timebase Accuracy

- ☐ Tick here if the variation from the 10 000 000ns is less than ±1 divisions.